

Day-2 Assessment

1. Primitive Data Types

Task: Create a program that accepts age, height, and weight of a person and prints them with appropriate data types.

Sample Input:

Age: 25

Height: 5.9

Weight: 68.5

Sample Output:

Age: 25

Height: 5.9

Weight: 68.5

package Daywise;

import java.util.Scanner;

public class Primitive {

public static void main(String[] args) {

Scanner sc=new Scanner(System.in);

System.out.print("Enter age: ");

int age=sc.nextInt();

System.out.print("Enter Height: ");

float height=sc.nextFloat();

System.out.print("Enter weight:");

double weight=sc.nextDouble();

System.out.println("\nAge: "+age);

System.out.println("Height: "+height);

System.out.println("weight: "+weight);

}

}

2. Variables

Task: Declare and initialize different types of variables to store a student's information: ID, name, marks, and grade. Print them.

Sample Input:

ID: 101

Name: Arun

Marks: 89.5

Grade: A

Sample Output:

Student ID: 101

Name: Arun

Marks: 89.5

Grade: A

package Daywise;

import java.util.Scanner;

public class Variable {

 public static void main(String[] args) {

 Scanner sc=new Scanner(System.in);

 int id=sc.nextInt();

 String name=sc.next();

 char grade=sc.next().charAt(0);

 double marks=sc.nextDouble();

 System.out.println("Student id:"+id);

 System.out.println("Name :"+name);

 System.out.println("Grade :"+grade);

 System.out.println("Marks :"+marks);

 }

}

3. Operators

Task: Accept two numbers and perform arithmetic, relational, and logical operations on them.

Sample Input:

Number1: 10

Number2: 20

Sample Output:

Addition: 30

Greater number: 20

Are both positive? true

package Daywise;

import java.util.Scanner;

public class Opera {

 public static void main(String[] args) {

 Scanner sc=new Scanner(System.in);

 System.out.print("Enter first num :");

 int n1=sc.nextInt();

 System.out.print("Enter sec num :");

 int n2=sc.nextInt();

 System.out.println("addition :"+(n1+n2));

 System.out.println("Greater num :"+(n1>n2?n1:n2));

 System.out.println("Are both positive :"+(n1>0 && n2>0));

 }

}

4. String Concatenation

Task: Create a greeting message using first name and last name entered by the user.

Sample Input:

First Name: Ravi

Last Name: Kumar

Sample Output:

Hello, Ravi Kumar! Welcome to the system

```
package Daywise;
```

```
import java.util.Scanner;
```

```
public class concat {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc=new Scanner(System.in);
```

```
        System.out.print("First name :");
```

```
        String first=sc.next();
```

```
        System.out.print("secound name :");
```

```
        String last=sc.next();
```

```
        System.out.println("Hello "+" "+first+" "+last+"! Welcome to the system");
```

```
    }
```

```
}
```

5. StringBuilder

Task: Accept a sentence and reverse it using StringBuilder.

Sample Input:

Input: Hello Java Learners

Sample Output:

Original: Hello Java Learners

Reversed: srenrael avaJ olleH

```

package Daywise;

import java.util.Scanner;

public class Reverse {

    public static void main(String[] args) {

        Scanner sc=new Scanner(System.in);

        System.out.println("input :");

        String input=sc.nextLine();

        StringBuilder sb=new StringBuilder(input);

        System.out.println("Original :"+input);

        System.out.println("reversed :"+sb.reverse());

    }

}

```

6. String API

Task: Count how many times a specific character appears in a string.

Sample Input:

String: banana

Character: a

Sample Output:

Character 'a' appears 3 times.

```

package Daywise;

import java.util.Scanner;

public class chcount {

    public static void main(String[] args) {

        Scanner sc=new Scanner(System.in);

        System.out.println("Enter String :");

        String str=sc.next();

        System.out.println("enter char to count");

        char ch=sc.next().charAt(0);
    }

}

```

```

int cou=0;
for(char c:str.toCharArray())
{
    if(c==ch)
    {
        cou++;
    }
}

System.out.println("character "+" "+ch+" "+"appears"+" "+cou+" "+"times")
}
}

```

8. Flow Control

Task: Based on a number entered, print whether it's positive, negative, or zero.

Sample Input:

Number: -5

Sample Output:

The number is negative.

```
package Daywise;
```

```
import java.util.Scanner;
```

```
public class Numcheck {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc=new Scanner(System.in);
```

```
        System.out.print("Enter number :");
```

```
        int n=sc.nextInt();
```

```
        if(n>0)
```

```
        {
```

```
            System.out.println("The number is posotive");
```

```
        }
```

```

else if(n<0)
{
    System.out.println("The number is negative");
}
else
{
    System.out.println("The number is zero");
}
}
}

```

9. Conditions

Task: Accept marks and display the grade using if-else.

Sample Input:

Marks: 76

Sample Output:

Grade: B

```
package Daywise;
```

```
import java.util.Scanner;
```

```
public class Grade {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc=new Scanner(System.in);
```

```
        System.out.print("Enter marks :");
```

```
        int marks=sc.nextInt();
```

```
        if(marks>=90)
```

```
            System.out.println("Grade : A");
```

```
        else if(marks>=80)
```

```
            System.out.println("Grade : B");
```

```
        else if(marks>=70)
```

```

        System.out.println("Grade : C");
    else if(marks>=60)
        System.out.println("Grade : D");
    else
        System.out.println("Grade : E");
    }
}

```

10. Switch

Task: Build a simple calculator using switch to perform operations (+, -, *, /).

Sample Input:

Number1: 10

Number2: 5

Operation: *

Sample Output:

Result: 50

```

package Daywise;

import java.util.Scanner;

public class Switch {

    public static void main(String[] args) {

        Scanner sc=new Scanner(System.in);

        System.out.println("Enter nu1 :");

        int n1=sc.nextInt();

        System.out.println("Enter num2 :");

        int n2=sc.nextInt();

        System.out.println("Enter operator((+,-,*,/) :)");

        char c=sc.next().charAt(0);

        switch(c)

        {

```



```
case '+':  
    System.out.println("res: "+(n1+n2));  
    break;  
case '-':  
    System.out.println("res :"+(n1-n2));  
    break;  
case '*':  
    System.out.println("res :"+(n1*n2));  
    break;  
case '/':  
    System.out.println("res :"+(n1/n2));  
    break;  
default:  
    System.out.println("Invalid operator ");  
}  
}  
}
```

11. Loops and Branching

Task: Print the first N even numbers using a loop.

Sample Input:

N = 5

Sample Output:

0 2 4 6 8

```
package Daywise;

import java.util.Scanner;

public class Even {

    public static void main(String[] args) {

        Scanner sc =new Scanner(System.in);

        System.out.print("Enter N:");

        int n=sc.nextInt();

        for(int i=0;i<n;i++)

        {

            System.out.print(2*i+" ");

        }

    }

}
```

12. Arrays

Task: Accept 5 numbers, store them in an array, and display their average.

Sample Input:

Numbers: 10, 20, 30, 40, 50

Sample Output:

Average: 30.0

```
package Daywise;

import java.util.Scanner;

public class Average {

    public static void main(String[] args) {

        Scanner sc=new Scanner(System.in);

        System.out.println("Enter size :");

        int n=sc.nextInt();

        int[] ar=new int[n];

        int sum=0;

        System.out.println("enter numbers ");

        for(int i=0;i<ar.length;i++)

        {

            ar[i]=sc.nextInt();

            sum=sum+ar[i];

        }

        System.out.println("Average :"+(sum/(ar.length)));

    }

}
```

13. Enum

Task: Create an enum for days of the week. Print a message depending on the day.

Sample Input:

Day: MONDAY

Sample Output:

Start of the work week!

```
package Daywise;

import java.util.Scanner;

public class Enum {

    enum Day{

        MONDAY,TUESDAY,WEDNESDAY,THURSDAY,FRIDAY,SATURDAY,SUNDAY

    }

    public static void main(String[] args) {

        Scanner sc=new Scanner(System.in);

        System.out.println("Enter day");

        Day day=Day.valueOf(sc.next().toUpperCase());

        switch(day)

        {

            case MONDAY:

                System.out.println("Start of the week");

                break;

            case FRIDAY:

                System.out.println("weekend");

                break;

            case SUNDAY:

                System.out.println("holiday");

                break;

            default:

                System.out.println("workday");

        }

    }

}
```

14. OOPs Concepts

Task: Create a Student class with fields for name and marks. Create an object and display its data.

Sample Input:

Name: Riya

Marks: 87

Sample Output:

Student Name: Riya

Marks: 87

```
package Daywise;
```

```
class student
```

```
{
```

```
    String name;
```

```
    int marks;
```

```
    void disp()
```

```
    {
```

```
        System.out.println("Student name :" +name);
```

```
        System.out.println("Marks: "+ marks);
```

```
    }
```

```
}
```

```
public class oops {
```

```
    public static void main(String[] args) {
```

```
        student s=new student();
```

```
        s.name="Vishnu";
```

```
        s.marks=452;
```

```
        s.disp();
```

```
    }
```

```
}
```

15. Inheritance

Task: Create a class Employee and a subclass Manager that extends Employee and adds department information.

Sample Input:

Name: Raj

Salary: 50000

Department: Sales

Sample Output:

Name: Raj

Salary: 50000

Department: Sales

package Daywise;

class Employee

{

 String name;

 double salary;

}

class manager extends Employee

{

 String dept;

 void disp()

 {

 System.out.println("Name :"+name);

 System.out.println("salary: "+salary);

 System.out.println("dept :"+dept);

 }

}

public class inherit {

```
public static void main(String[] args) {  
    manager m=new manager();  
    m.name="Raj";  
    m.salary=12000;  
    m.dept="sales";  
    m.disp();  
}  
}
```